

Department of Mechanical Engineering

Course Outcomes of all courses of B Tech 8th semester MECHANICAL

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C411 Robotics D037811(037)	C 411.1	Demonstrate the basic knowledge of terminologies, characteristics, components and applications of robotics systems. (Level 3)
	C 411.2	Apply spatial transformation to obtain forward kinematics equation of robot manipulators. (Level 3)
	C 411.3	Perform position analysis and velocity analysis of direct and inverse kinematic models of robots. (Level 3)
	C 411.4	Describe sensing technologies and robotics vision system and choose the appropriate for a given application. (Level 2)
	C 411.5	Develop and analyze the mathematical model for motion planning and control of robot manipulators and describe robotics applications. (Level 6)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C412- Automobile Engineering D037831(037)	C412.1	Describe the basic structure of an automobile with applied engineering principle in its design. (Level 2)
	C412.2	Describe clutches and fluid flywheel and solve related problems. (Level 2)
	C412.3	Describe construction and working principle of gear box and torque converter and analyze problem related automobile performance. (Level 2)
	C412.4	Describe construction and working of propeller shaft, differential, axle assembly, tyres and braking system of an automobile (Level 2)
	C412.5	Describe construction and working of steering and electrical systems with applied engineering principle in its design. (Level 2)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C413- Production & Product Management D000806(037)	C413.1	Explain the product development process of a new product. (Level 2)
	C413.2	Explain product design methods. (Level 2)
	C413.3	Explain the concepts of design for manufacture. (Level 2)
	C413.4	Explain the concepts of Industrial design. (Level 2)
	C413.5	Discuss legal issue pertaining to product design and management of product development projects. (Level 5)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C414-Robotics (Lab) D037821(037)	C414.1	Demonstrate the working of different types of robot (Level 3)
	C414.2	Demonstrate the knowledge of the robotic system design, sensors, actuators, vision systems and robotic application. (Level 3)
	C414.3	Design , model and analyze gripper (Level 6)
	C414.4	Write the program for linear and nonlinear trajectories and forward kinematic problems by using software. (Level 5,6)
	C 414.5	Develop virtual model for kinematic and dynamic verification of robotics structure using software (Level 6)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C415- Computer Aided Simulation & Analysis (Lab) D037822(037)	C415.1	Model and analyze structural problem using commercial simulation and analysis software. (Level 4)
	C415.2	Model and analyze Thermal problem using commercial simulation and analysis software. (Level 4)
	C415.3	Model and analyze Fluid Flow problem using commercial simulation and analysis software. (Level 4)
	C415.4	Analyze Coupled field problems using commercial simulation and analysis software. (Level 4)
	C415.5	Determine the natural frequency of an object using commercial simulation and analysis software. (Level 5)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C416- Project (Phase-II) D037823(037)	C416.1	Recognize the need and identify the problem related to industry and society through literature and environment focusing on practical conditions. (Level1)
	C416.2	Develop and select a solution to identified problem in a cost effective manner. (level 3, 5)
	C416.3	function in a team and adapt as per requirement to achieve desired goal with ethical practices (level 4, 6)
	C416.4	Apply principles to solve problems and interpret the result. (level 3, 5,)
	C416.5	Relate the impact of engineering solutions in society and classify modern tools .(level 2,3)